RUTGERS HEALTH
School of Health Professions
Clinical Laboratory and Medical Imaging Sciences

Department of Clinical Laboratory and Medical Imaging Sciences
Medical Laboratory Science Program
MLSC 4329 Hematology II
(Last Revision August 26, 2026)

COURSE DESCRIPTION

Instruction and student laboratory practice focus on special procedures in hematology and hemostasis with emphasis on problem-solving. Selected topics include advanced concepts in the physiology of hematopoiesis and hemostasis, and clinical correlation of data in conditions such as anemia, leukemia, and hemorrhagic and thrombotic disorders.

Credits: Three (3)

Hours, Course Format, and Modes of Instruction

The course is designed as a hybrid learning model and will be delivered online via Zoom, the Canvas Learning Management System, and in-person. The course is scheduled once a week for 16 weeks throughout the Fall semester. The course includes online & campus lectures/learning activities, online & campus laboratory sessions and simulations, and mandatory review sessions. Pre-recorded lectures and mini lectures will be available on Canvas for selected units and lectures. It is expected that the student will devote 135 hours for the course or an average of 8.5 hours per week to include reading online units, reading assigned textbook chapters, visiting assigned websites, watching assigned videos, preparing for and attending online and on-campus laboratory sessions, attending review sessions, completing assignments and activities, and independent study for exams.

Prerequisites/Corequisites

MLSC 2129 Hematology I

Course Coordinator and Faculty

Course Coordinator and Primary Instructor
Stephanie B. Cochrane, EdD, MLS (ASCP)^{CM}
Assistant Professor
sbn26@shp.rutgers.edu, 973-972-3298 (SHP Newark Office), 910-273-0104 (mobile)

Additional Laboratory Faculty:
Mr. Steve Bellows, MS, MT (ASCP)
Ms. Zareen Abed, MLS(ASCP)

Course Goals

Upon completion of lecture, online and student laboratory activities, you will be able to achieve the course goals listed below:

A. Assess, analyze, implement, and evaluate laboratory tests and results, incorporating pre-analytical, analytical, and post-analytical measures of quality assurance and developing solutions to problems concerning the generation of laboratory data.

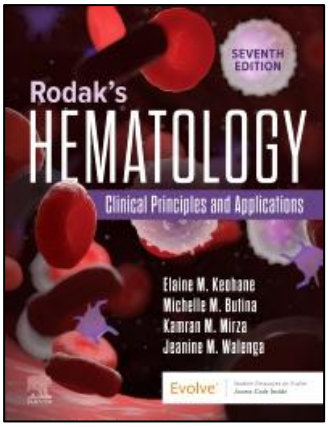
In addition, the student will be able to achieve the course goals listed below:

1. For special procedures in hematology and hemostasis, you will be able to:
    - Explain the principle, procedure, interpretation, and clinical application.
    - State appropriate units, sources of error, reference ranges and specimen requirements.
    - Correctly perform calculations if necessary.
    - Identify abnormal and critical values.
2. Given a problem or set of laboratory data in hematology and hemostasis, you will be able to:
    - Correlate results with other laboratory data and quality control results.
    - Correlate results with normal and abnormal physiologic conditions.
    - Determine an appropriate course of action and/or confirmatory testing.
3. Given selected peripheral blood and bone marrow smears or electronic images, you will be able to appropriately evaluate them by:
    - Correctly identifying normal, immature, and abnormal blood cells.
    - Suggesting possible conditions associated with the blood picture.
    - Suggesting appropriate confirmatory testing
4. Given the appropriate supplies, reagents, and equipment, you will be able to perform selected laboratory procedures under the conditions and criteria specified in the Student Laboratory Manual.

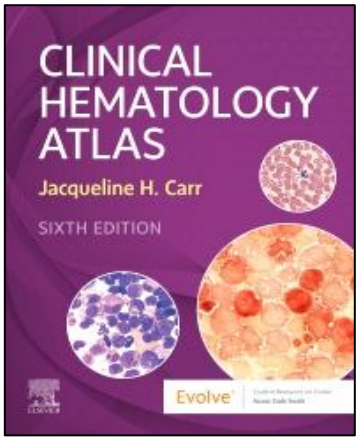
During the course, you will be able to demonstrate the behaviors listed in the affective objectives according to the specified criteria.

Textbook, Readings, and Visual Aids

Optional Purchase



Required Purchase



1. Keohane, Otto, Butina, Mirza, & Walenga (Eds.). Rodak's Hematology: Clinical Principles and Applications, 7th Ed. Elsevier 2024. Paperback ISBN: 9780323936507
2. Carr, J: Clinical Hematology Atlas. 6th edition. Elsevier Health Sciences, 2022. ISBN: 9780323711920

Please purchase your books through Barnes and Noble (http://www.barnesandnoble.com/) or Amazon.com (http://www.amazon.com/). Please note: if you are a Barnes and Noble member ($25.00 per year), you will get the member rate of all textbooks.

Available Reference Books:

- Carr, Rodak. Clinical Hematology Atlas. 3rd edition. Elsevier Health Services, 2009. ISBN 978-1-4160-5039-1
- Rodak, Fritsma, Doig. Hematology, Clinical Principles and Applications, 3rd edition. Missouri, Saunders, 2007.
- McKenzie, SB. Clinical Laboratory Hematology. New Jersey: Prentice Hall, 2004. ISBN # 013-019996-6
- Diggs LW, Strum D, Bell A. Morphology of Human Blood Cells. Latest Edition. Chicago, Abbott Laboratories.
- Harmening DH (ed). Clinical Hematology and Fundamentals of Hemostasis. 4th edition. Philadelphia, Davis, 2002.

Helpful Internet Sites

- American Society for Hematology (ASH Image Bank: http://imagebank.hematology.org)
- For Hemostasis: The Fritsma Factor: Audio Modules (Lectures on various topics in Hemostasis): http://www.fritsmafactor.com/newfritsmafactor/?page_id=997
- Testing.com (Lab Tests Online): https://www.testing.com/

Attendance

Attendance at lecture, review and online microscopy lab sessions is mandatory. It is very difficult to make-up on-campus lab sessions if you are absent. Students are expected to be on time for all class sessions (Zoom and/or in-person) since announcements and explanations of the day's activities occur at the beginning of each session.

Excessive absenteeism or tardiness (as defined in the MLS Student Policy Manual) will result in an academic warning and may jeopardize your satisfactory completion of the course.

If you are unable to attend class or participate online, please call email the course coordinator, Dr. Stephanie Cochrane in course e-mail before the class session.

Requirements and Evaluation

The course requirements and evaluation is based upon achievement of the course objectives. The course requirements and their % weight in the determination of the final grade are as follows:

In order to pass the course, you must:

1. Achieve a minimum final course grade of 73%.
2. Achieve a minimum of 70% on all theoretical course exams.
3. Achieve a minimum of 80% on the practical exam.
4. Achieve a pass grade on all lab reports, problem cases, and assignments.
5. Have acceptable attendance at class, lab sessions, and online.

Additionally, the lab includes components which are considered non-negotiable essentials for clinical practice. The ability to successfully complete these tasks or master these concepts is required to pass the course, regardless of your numerical or letter grade. Determination of competency will be made by the instructor and/or the Program Director. Non-negotiable components are listed below:

- Recognize and correctly identify normal and abnormal cells on peripheral blood slides
- Consistently count differentials correctly and in a timely manner

Course Requirements and % Weight

Requirement	Weight
Exam 1 (Unit 1)	20%
Exam 2 (Units 2 - 3)	20%
Cumulative Final Exam (Unit 1 – 5, excluding Unit 4)	35%
Laboratory Practical Exam	15%
Hematology Diagnostic Console AI Chatbot Assignment (Total of 3)	5%
Online Microscopy Labs MediaLab, Inc. (LabCE) Case Simulator CellaVision Proficiency Software	5%
Lab Reports **Study Questions/PSAs/Case Studies** **HARL activities** **Formative Quizzes and Unit Self-Assessments**	Pass/Fail

Absence from Exams

It is your responsibility to be present for all scheduled examinations. If you are absent from a scheduled examination, you will receive a grade of "F" for that exam unless the absence is due to illness or a sudden unexpected emergency that can be documented in writing by a physician or other acceptable source as determined by the program director and course instructor. This evidence must be submitted upon your return to class.

If you miss an exam due to acceptable circumstances above, you will be examined at the convenience of the course coordinator within 2 weeks of your return. A different form of the

exam may be utilized. Different versions of exam may also be given (e.g., Exam A, Exam B, etc.)

Special Requirements:

Since a portion of this course is web-enhanced, you will need reliable access to the Internet through an Internet Service Provider. You also need to have the computer hardware, software, and browser requirements for Canvas.

Students assume the cost of printing online unit materials for the course.

Evaluation and Grading

Evaluation and Assessment Methods

Lecture Exams: Three (3) theoretical exams will be administered in-person, are proctored, and are "closed-book". Theoretical exams are almost exclusively multiple-choice questions. Exam 1 covers Unit 1, Exam 2 covers Unit 2 and 3, and the Cumulative Final Exam covers Units 1 – 5, excluding Unit 4. Exam 1, Exam 2, and the Cumulative Final Exam account for 20%, 20%, and 35% of the final course grade, respectively.

Laboratory Practical Exam: The practical exam consists of electronic images from cases in which you must correctly identify normal, immature, and abnormal cells, suggest possible conditions for the cases, and suggest appropriate confirmatory or follow-up lab tests. The practical exam is administered on campus. Students must pass the practical exam with a minimum score of 80% or a B- to pass the course. The practical exam accounts for 15% of the final course grade.

Hematology Diagnostic Console AI Chatbot Assignment: In this interactive assignment, students will engage with an AI-powered diagnostic simulation to interpret complex hematologic case data. Acting as medical laboratory science student completing a hematology clinical practicum at RUHS Hospital, students will analyze patient case files via a virtual console (ChatGPT), request lab values (e.g., CBC, peripheral smear), and build diagnostic reasoning through inquiry and reflection. The simulation promotes the application of ASCP-level terminology, critical thinking, and clinical decision-making. Students will submit a transcript or summary of their diagnostic process, a summary table of key findings and interpretation, a 150-word reflection, and feedback provided by the AI. There is a total of 3 AI Chatbot Assignments that will account for 5% of the final course grade.

Online Microscopy Labs (OMLs):

***WBC Differential Case Simulator*:** The WBC Differential Case Simulator is a great resource for practicing the identification of normal and abnormal WBCs. The LabCE White Blood Cell Differential Simulator includes 25 expert-reviewed differentials each with 100 slide images. You will use the the MediaLab, Inc. (LabCE) case simulators as practice and to better prepare you for the on-campus student laboratory sessions. You will access the site by clicking on this link: https://www.medialab.com.

If not already provided in MLSC 2129 Hematology I, the instructor will provide students with their username and password. You will log in and start doing the assigned cases. At the end of each case, you will find your results and/or final grade as well as the correct answers. Also, a video at the end explains why those answers were chosen.

***CellaVision Proficiency Software*:** The CellaVision Proficiency Software is designed to help promote competency in the area of cell morphology. The instructor will create test scenarios of various digital slides. Students will perform a WBC classification and RBC characterization on one or more digital slides. Upon completion, students will be able to review their results and compare their differential to that of the instructor.

Completion of Online Microscopy Labs are a course requirement and account for 5% of the final course grade.

Lab Reports: Lab reports may be required to document students' work during on-campus laboratory sessions to access performance and evaluation of results and examining peripheral blood smear images and/or prepared slides. Unless instructed otherwise, students must complete the lab reports independently and must not consult with other students about the reports. Requirements for each lab reports will be detailed by the instructor prior to the laboratory session. The lab reports are graded as PASS/FAIL.

Study Questions: Study questions are provided for each unit. They use a short answer format and are given to help clarify course material and emphasize important concepts. Study questions are graded as PASS/FAIL.

Problem-Solving Activities (PSA): Problem-solving activities are to aid the student in developing higher-order thinking skills regarding the evaluation, analysis, and interpretation of laboratory data. Students are encouraged to complete PSAs after each lesson to review the concepts learned. PSAs include case studies where students must determine the most likely patient diagnosis or recommend appropriate follow-up tests and hematological calculations. The instructor will provide feedback for each PSA prior to course examinations. PSAs are graded as PASS/FAIL.

Case Studies: Case Studies are provided (Hemostasis, RBCs, and WBCs) as part of the Study Skills Guides for each exam. They're given to help students correlate clinical manifestations, laboratory test results and morphology, determine possible conditions, and suggest appropriate follow-up testing. Case studies are graded as PASS/FAIL.

Hematology Applied Reasoning Lab (HARL) activities: The Hematology Applied Reasoning Lab (HARL) is a structured, collaborative learning experience designed to strengthen diagnostic reasoning in laboratory medicine. Through various learning activities integrated throughout each lesson/unit (e.g., Pattern Pause, Highest-Yield Clue, Decode the Smear, etc.) students will practice interpreting CBC data, evaluating peripheral smear findings, building differential diagnoses, and strategically analyzing complex questions. These sessions emphasize evidence-based thinking, professional discussion, and reflection, helping students connect lecture concepts to clinical problem solving while building confidence in their diagnostic decision-making. For more details on HARL, see the Canvas course page, *How This Course Works*. HARL activities are graded as PASS/FAIL.

Formative Quizzes: Short formative quizzes (10 – 15 questions) are provided for each part of the unit to help retrieve and reinforce essential concepts before applying them to more complex problems and cases. Each quiz focuses on key terminology, laboratory findings, disease characteristics, and high-yield patterns from the corresponding lesson. These low-stakes assessments are designed to identify knowledge gaps, strengthen long-term retention, and prepare you for higher-level diagnostic reasoning activities and course assessments. Formative quizzes are graded as PASS/FAIL.

Unit Self-Assessments: Unit Self-Assessments are provided at the end of each unit to aid in retrieval practice and self-testing in preparation for exams. They utilize multiple formats (i.e., multiple choice, matching, drag and drop, etc.) and are given to help clarify course material and emphasize important concepts. You can access unit self-assessments from the course in Canvas under each Unit Wrap Up. There are approximately 20-30 questions per unit self-assessment. Unit Self-assessments are graded as PASS/FAIL.

Affective Evaluation Checklist: Each semester, course faculty complete a combined affective evaluation of your behaviors and attitudes demonstrated in the theoretical courses in which you are enrolled. Achievement of these objectives is very important for you as a student to develop into a competent and professional medical laboratory scientist. Although the affective evaluation is not graded, you are expected to achieve a minimum rating of "meets expectations" in each category. Faculty use the affective evaluation to 1) formally recognize students whose behaviors exceed expectations, 2) provide positive feedback to students whose behaviors meet expectations, and 3) alert and counsel students whose behaviors need improvement.

Feedback on Progress: Students will receive feedback on performance after completion of exams and during lab and seminar sessions.

Grade Determination

Letter grades will be assigned according to the following scale:

Final Letter Grade	Weighted Average of All Requirements
A	93.0 – 100
A-	90.0 – 92.9
B+	87.0 – 89.9
B	83.0 – 86.9
B-	80.0 – 82.9
C+	77.0 – 79.9
C	73.0 – 76.9
C-	70.0 – 72.9
D+	67.0 – 69.9
D	63.0 – 66.9
F	<63

Unsatisfactory Performance: You will receive a written warning and suggestions for improvement if your performance is unsatisfactory (average below 73%) at mid-course. You are allowed one make-up for the student laboratory practical exam if your grade is below 80%.

Passing the lab practical exam with at least a grade of 80% is required to pass the course.

If you receive a "Needs Improvement" rating on the Affective Evaluation, you are required to meet with the Program Director and other applicable faculty for counseling and discussion of strategies for improvement. A midsemester affective evaluation may be given if a problem is identified early in the semester. After receipt of a "Needs Improvement" rating and participation in counseling session(s), a student is expected to implement strategies to improve his/her behavior/attitude by the next evaluation. A student who does not show improvement may be referred to the Department Committee on Admissions and Academic Performance for review and appropriate action.

Please refer to the MLS Program Student Policies for further information on grading, unsatisfactory performance, and make-up policies.

Recording Devices: When personally owned communication/recording devices are used by students to record lectures and/or classroom lessons, such use must be authorized by the faculty member or instructor who must give either oral or written permission prior to the start of the semester and identify restrictions, if any, on the use of mobile communication or recording devices.

Laboratory Dress Code

The student will be expected to attend class clean and neatly dressed in long pants or scrubs and wear closed-toe shoes. A disposable laboratory coat must be worn snapped or buttoned up during all laboratory sessions. Hair that is shoulder length or longer must be worn up or securely tied back. Gloves must be worn when handling biological materials.

Course Evaluation

Students will have an opportunity to evaluate the course and course faculty at the end of the course. The faculty encourages students to provide constructive feedback on course and instructor evaluations. Student feedback is valuable in structuring the course and providing a meaningful learning experience.

CLS Code of Conduct and Professional Behavior

All students in the Department of Clinical Laboratory and Medical Imaging Sciences (Doctoral, Masters, and Bachelors) are expected to maintain a high level of professionalism and respect in interactions with faculty. Students can access the Code of Conduct from the Canvas course Welcome module.

Faculty/Student Honor Code

Refer to your SHP Student Handbook.

The faculty of Rutgers-School of Health Professions believe that students must observe and support high standards of honesty and integrity in all aspects of education, practice, and research. For this reason, all matriculated and non-matriculated students in this course are

expected to abide by the School's Faculty/Student Honor Code and accept responsibility to help ensure that these standards are maintained by reporting violations of the Honor Code observed in others. All violations will be considered with gravest concern and may be punishable with sanctions as severe as suspension or dismissal.

All written work must be in compliance with the Faculty/Student Honor Code and must be in your own words.

- Copying/pasting content verbatim from other sources including course materials in discussions, written assignments, exams and other assessments is considered plagiarism unless it is in quotes and cited. Note that quotations are not typically used in scientific papers.
- Copying/pasting substantial content from a source so that it follows the thought process, organization and flow of information is also considered plagiarism, even if it is paraphrased and referenced.
- Content in assignments and exams must be written in your own words, and the ideas and concepts presented must be appropriately cited unless they are considered common knowledge

Use of Artificial Intelligence (AI) Software

SHP faculty are aware of digital platforms and artificial intelligence (AI) software that can assist students in preparation of their assignments. Course faculty may authorize student use of AI software for certain circumstances. Unauthorized use of AI (including but not limited to any version of ChatGPT) constitutes a violation of academic integrity and the SHP Student Honor Code. Even with authorized use, students should be aware that using AI may produce work that is missing information or not correct. Students are responsible for the finished product they produce, and the sources used to generate that work.

For all issues regarding academic integrity, refer to SHP Disciplinary Policies and Procedures: https://shp.rutgers.edu/disciplinary-process/

Program Notification of Extenuating Circumstances: Student Responsibilities

Occasionally circumstances arise which are beyond the students control and which may affect the student's ability to perform on a particular examination, clinical experience, assignment, or other grading encounter. SHP programs are supportive of and invested in their students, and will help them simultaneously navigate the required program exercises and deal with the exigent circumstances to the best of their ability. However, as stated in SHP Academic Policy 3.1.1 Academic Decision and Grade Review, it is the responsibility of the student to notify the program or faculty member of the exigent circumstance **by email, and before** the examination or graded exercise begins. Notification by the student after the examination or graded exercise disallows that particular circumstance from being considered as a factor in the student's performance, or in the determination of the student's grade. For a circumstance to be considered by a program, it must have been notified of that circumstance by the student before the graded exercise begins, and the notification must be by email. In that way, there is no question regarding what the student claimed and when that claim was made.

Services Available to Students

A variety of student services are available through Rutgers-SHP. Please note that sometimes it is hard for a student to determine what kind of services or supports are needed. Review the

information below; if you have any questions about student services at Rutgers-SHP contact your Program Director who will help you access the appropriate services.

General Student Services

The Rutgers-SHP Office of Student Affairs (OSA) is devoted to providing support services that will assist students in their pursuit of ensuring personal and academic success. A variety of services are offered and extended to all SHP students. They include but are not limited to academic tutoring, college survival tips, counseling services, the Educational Opportunity Fund (EOF), and services for students with disabilities. For more information go to the SHP OSA webpage found https://shp.rutgers.edu/student-affairs/.

Services for Students with Disabilities

Rutgers University is committed to the creation of an inclusive and safe learning environment for all students and welcomes disabled students/students with disabilities into all the University's educational programs. The Office of Disability Services (ODS) is responsible for the determination of appropriate accommodations for students who encounter barriers due to disability. Once a student has completed the ODS process (registration, initial appointment, and submitted documentation) and reasonable accommodations are determined to be necessary and appropriate, a Letter of Accommodation (LOA) can be requested and will be sent to the student and instructor. This should be done as early in the semester as possible as accommodations are not retroactive, and a discussion with instructors should occur about how the accommodations will be implemented. More information can be found at https://oasa.rbhs.rutgers.edu/office-of-disability-services/ or by contacting the office of disability services at odsrbhs@rbhs.rutgers.edu, or call Jenna Rose at 973-634-4823/Cindy Poore-Pariseau at 973-640-0058.

Supported Education Services

Educational support and services for students with mental health conditions are also available through a Supported Education Program at Bridgeway Behavioral Health Services. To gain access to these specialized services, contact Bridgeway Career Services at 908-687-9666.

Student Wellness Services

The Student Wellness Program is available to any student who might need emergency mental health services or to speak to a counselor. Drop-in times are scheduled, and appointments can be set up. To find out about drop-in times or to set up an appointment please contact one of the following:

Rutgers-University Behavioral HealthCare
Employee Assistance/Student Wellness Program
183 South Orange Avenue, Newark
973-972-5429

Rutgers-University Behavioral HealthCare
Employee Assistance/Student Wellness Program
242 Old New Brunswick Road, Piscataway
732-235-5933

Rutgers-University Behavioral HealthCare
Employee Assistance/Student Wellness Program
1 Echelon Plaza, Suite 101
227 Laurel Road, Voorhees
856-770-5750

Student Food Pantry

The RBHS Food Pantry is available to all students, faculty, and staff experiencing food insecurity. The pantry is located at the Stanley S. Bergen Building (SSB), Rm GB-01 at 65 Bergen Street in Newark, New Jersey. To find out about how to order food and when to pick it up, please view the website https://rutgershealth.org/rbhs-food-pantry-newark .

Psychomotor Objectives (Student Laboratory):

The student laboratory manual contains the psychomotor objectives of the course. After performance of the psychomotor objectives in each lab exercise, students must submit lab reports of their work. The conditions and criteria for the psychomotor objectives are specified in the student lab manual under each laboratory exercise. Students must perform the procedures in each exercise until the results meet the specified criteria in order to get a "P" (Pass) grade on the lab report.

Achievement of the blood cell identification objectives will be met when the student scores a minimum of 80% on a "closed-book", proctored, practical examination consisting of 25 cell identifications on microscopic slides or electronic images.

Affective Objectives:

This section contains the affective objectives of the course.

Achievement of the **Essential Affective Objectives** will be met when the student demonstrates the specified behaviors at least 100% of the time.

Achievement of the **Additional Affective Objectives** will be met when the student receives a "Meets Expectations" evaluation on the specified behaviors. These behaviors are evaluated as "Meets Expectations", "Exceeds Expectations", or "Needs Improvement" as follows:

- Exceeds Expectations: The student demonstrates the specific behavior/attitude 100% of the time, and his/her actions consistently exceed expectations of a student in the Program.
- Meets Expectations: The student demonstrates the specific behavior/attitude at least 80-99% of the time.
- Needs Improvement: The student demonstrates the specific behavior/attitude less than 80% of the time

Achievement of the **Affective Objectives that are encouraged** is not a requirement for passing the course.

See section on Unsatisfactory Performance in the syllabus for a description of actions taken for a "Needs Improvement" evaluation.

Communication

1. Appropriately follow oral directions.
2. Appropriately follow written directions.
3. Ask appropriate questions.
4. Appropriately respond to questions.
5. Demonstrate effective and appropriate oral communication.
6. Demonstrate effective and appropriate written communication.
7. Check and use UMDNJ and Moodle course e-mail on a regular basis.

Cooperation

1. Demonstrate courtesy in contacts with others.
2. Willingly perform tasks required of a student.
3. Treat equipment with care.
4. Use lab supplies and reagents carefully without unnecessary waste.
5. Work effectively with others.
6. Help other students when appropriate.

Self Confidence/Self Reliance

1. Perform lab procedures without excessive repetition.
2. Demonstrate an appropriate level of confidence (e.g. do not require unnecessary assurances from instructor; not overconfident).
3. Recognize lab results or course situations that should be brought to the instructor's attention.
4. Maintain composure under stress.
5. Be able to work independently.

Organization Ability

1. Maintain a clean and orderly work area.
2. Perform tasks with minimal disruption of others.
3. Follow the flow of tasks in the student lab.

Interest/Initiative/Motivation

1. Be prepared for class activities (e.g. read materials before lectures, labs, etc.).
2. Leave class at the scheduled time.
3. Demonstrate interest in work and eagerness to learn.
4. Begin activities on own initiative.
5. Correlate theoretical with student lab activities (know what you are doing and why you are doing it).
6. Actively participate in class, online, and student lab activities.

Effectiveness/Efficiency/Response to Work

1. Submit assignments on time.
2. Work systematically and efficiently.
3. Follow through on assigned tasks.
4. Work under expected time/volume pressure with minimal loss of accuracy.
5. Respond promptly and appropriately in most situations.
6. Concentrate on task at hand.

Professional Decorum/Commitment

1. Be appropriately attired and groomed (see MLS Program Policies and Safety Manual)
2. Demonstrate speech and manner appropriate for professional and educational setting.
3. Participate in professional activities.
4. Demonstrate respect and commitment to the profession.
5. Comply with policies regarding the use of electronic devices during class and lab sessions.

Additional Affective Objectives that are encouraged

1. Volunteer for committees, open houses, and off-campus school events.
2. Assume leadership roles in campus and professional settings.
3. Join The American Society for Clinical Laboratory Sciences of New Jersey.

Prepared by Elaine M. Keohane, Ph.D., revised 2015, 2016; Ruth Perez, 2018; Stephanie Cochrane, EdD, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026.